

 Meeting · June 2026

IPDO - Aurigene Collaboration POC

Capabilities & Computational Formulation Workflows

 Confidential

 IPDO – Aurigene Collaboration POC

Capabilities

Characterization

Crystal Structure

- Polymorphic purity; Crystal Structure Determination from Powder-XRD
- Salt or Co-crystal; impact of temperature on proton position
- Phase Composition Quantification (multiphase XRD profile fitting)

Morphology

- Crystal habit & dominant growth face prediction; surface chemistry relevant to processing

CSP

- Global lattice energy minimization; ranking of plausible crystal structures
- Simulated PXRD; hydrogen-bonding analysis

Spectroscopy

- IR, Raman, SS-NMR, microED, crNMR; TEM / SEM image analysis

Image Analysis

Pre-Formulation & Formulation

Solubility

- Solubility profiling

Screening Panel

- Screen for Solvates, Salts, Co-crystals, ASDs

ASD

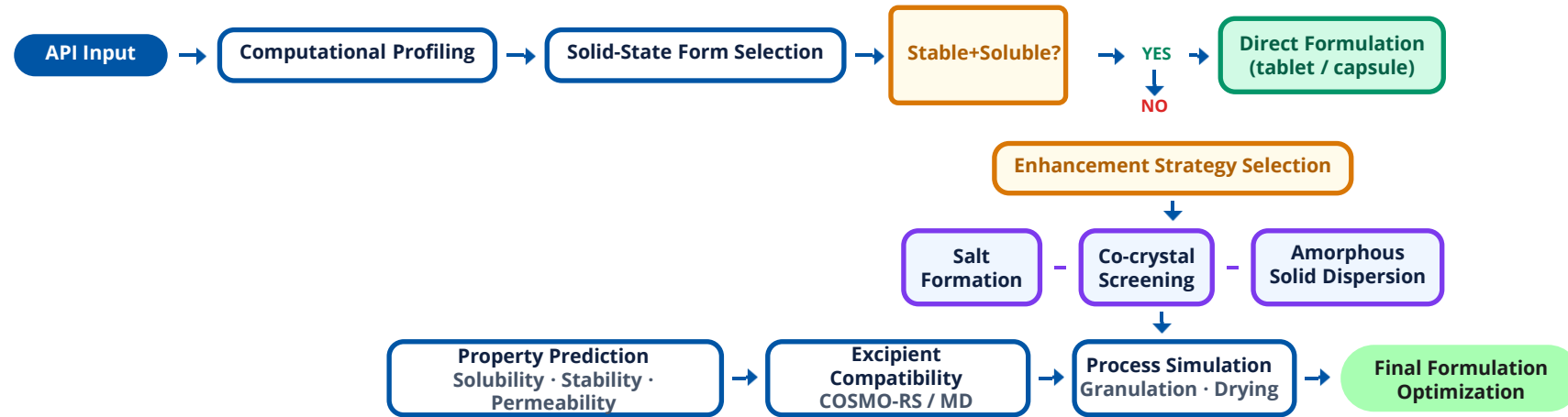
- Miscibility, Stability (Glass Transition Temp)
- Separation & Dissolution, Hygroscopicity

Excipient Science

- Excipient screening; API solubility and LogP in excipient
- Solubility parameters; API-excipient mixing enthalpy

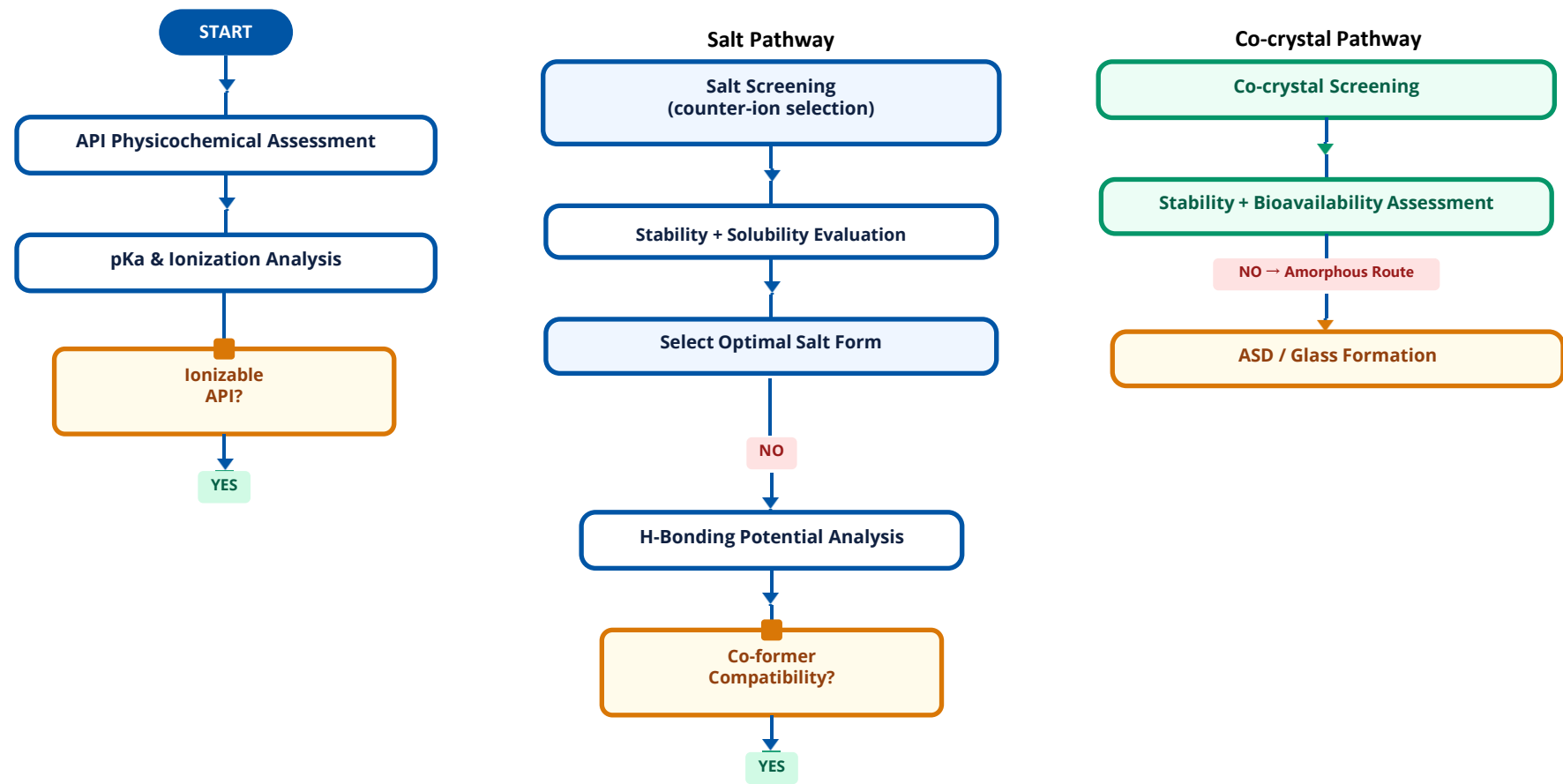
Integrated Formulation Decision Workflow

A complete end-to-end computational formulation pipeline — from API profiling through solid-state selection, enhancement strategy branching, property prediction, excipient modeling, and process simulation to final optimized formulation.



Solid-State Form Selection

Systematic decision logic for selecting the optimal solid-state API form — salt, co-crystal, or amorphous — driven by ionizability and hydrogen-bonding potential.



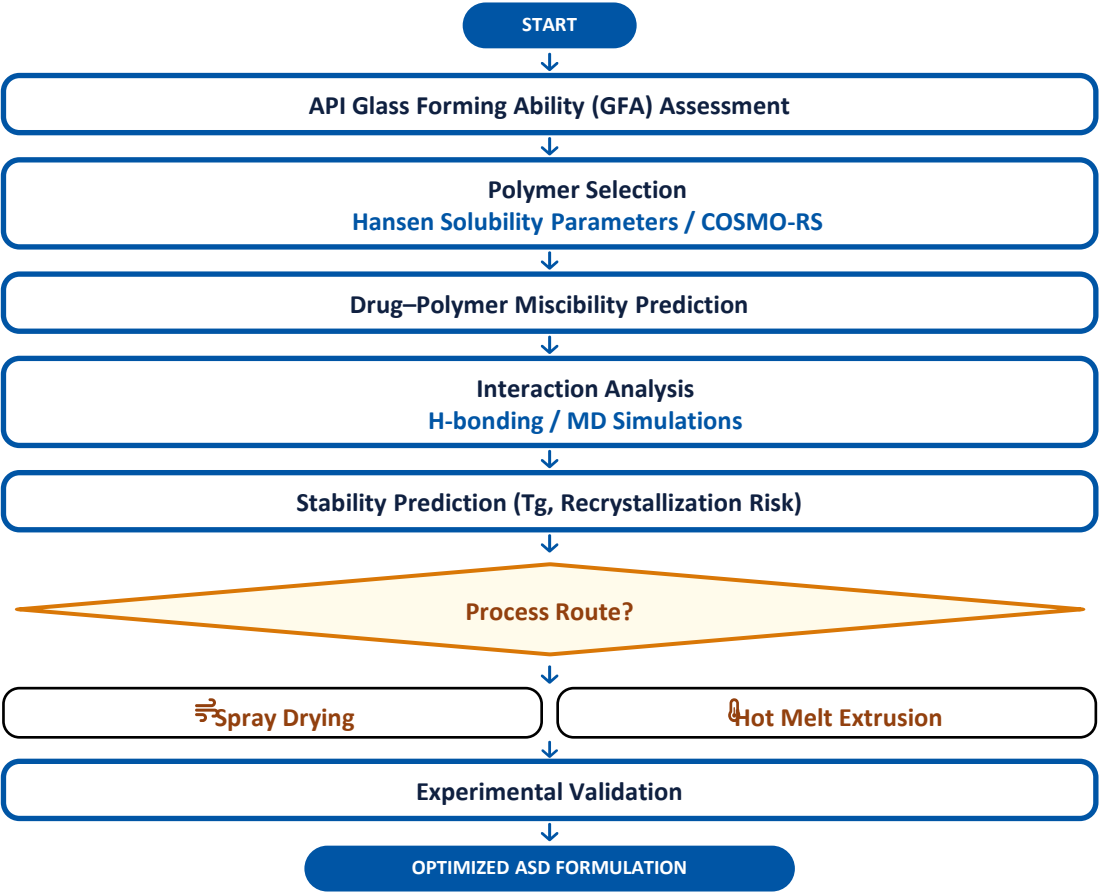
● Salt — ionizable APIs with suitable counter-ions

● Co-crystal — non-ionizable with H-bond co-formers

● Amorphous (ASD) — no viable counter-ion or co-former

Amorphous Solid Dispersion (ASD) Design

A structured computational workflow — from glass forming ability through polymer selection, miscibility prediction, and process route selection to validated ASD output.



Key Computational Methods



GFA Assessment

Evaluates API tendency to form stable amorphous glass. Informed by Tg, fragility index, and crystallization kinetics.

Entry Gate



Polymer Selection

Hansen solubility parameters and COSMO-RS thermodynamics guide optimal polymer screening for drug–polymer compatibility.

COSMO-RS / HSP



Miscibility & Interaction

MD simulations and quantum mechanics validate H-bonding potential. Flory-Huggins χ predicts thermodynamic miscibility.

MD / QM



Stability Modeling

Tg prediction and recrystallization risk are critical indicators for long-term ASD physical stability and shelf life.

Critical Output

Critical Stability Indicators

Tg Prediction

Glass transition temperature confirms amorphous state retention above storage conditions

Recrystallization Risk

Modeling predicts propensity for drug nucleation and crystal growth within the polymer matrix



Optimized ASD Formulation

Computationally screened polymer, validated miscibility, experimentally confirmed stability

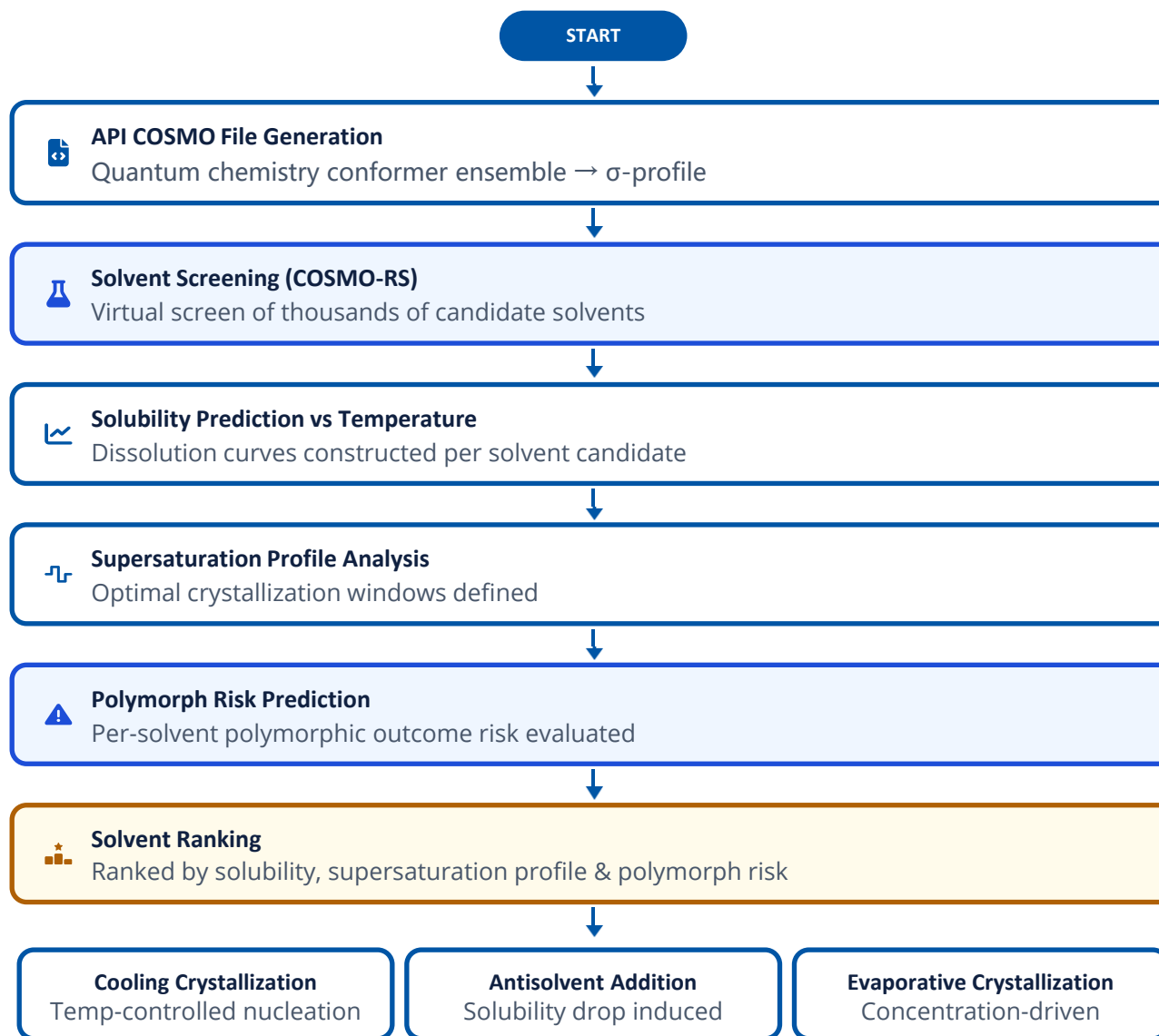
COSMO-RS Driven

Solvent Selection for Crystallization

COSMO-RS-driven solvent screening delivers ranked solvent candidates with solubility-temperature profiles, supersaturation analysis, and polymorph risk assessment to guide crystallization process design.

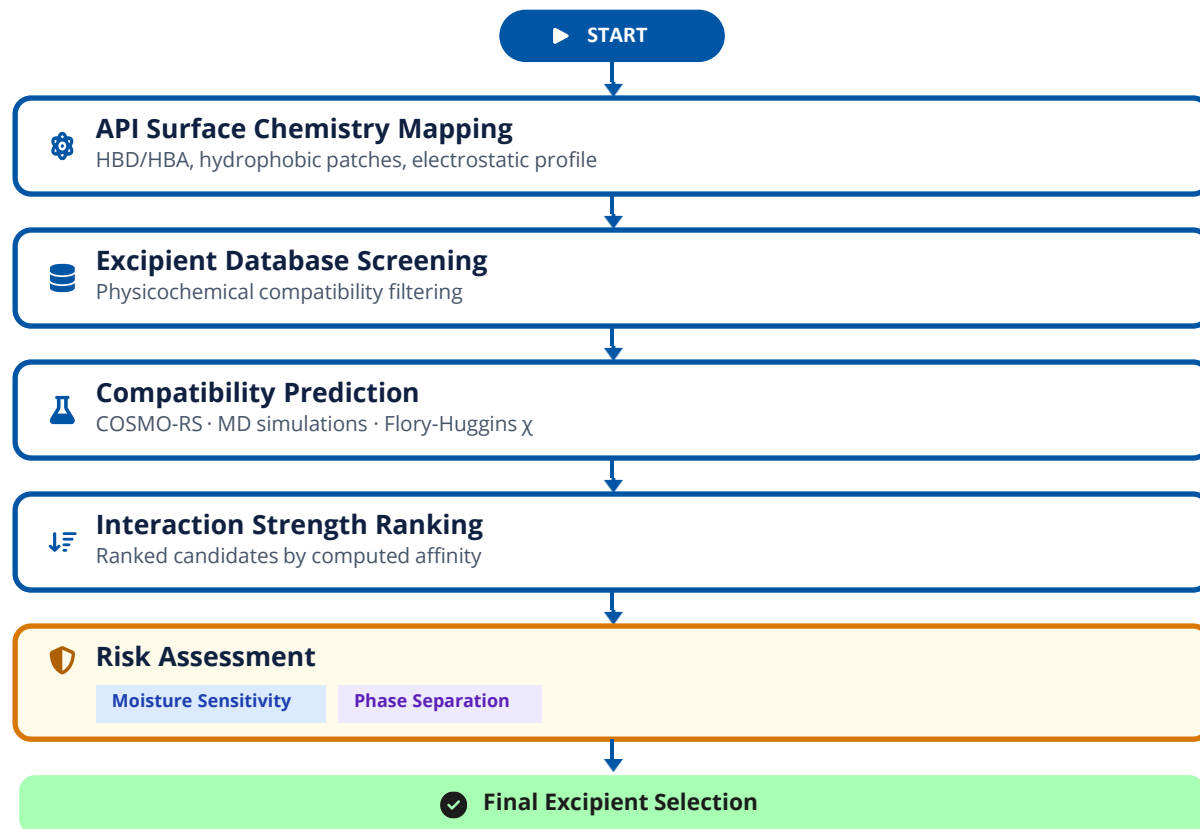
Key Insight

COSMO-RS enables rapid virtual screening of thousands of solvents; supersaturation profiles predict crystallization windows.



Excipient Selection Workflow (Computational)

Computational excipient selection maps API surface chemistry, screens databases for compatibility, predicts interaction strength via COSMO-RS / MD / Flory-Huggins, and risk-assesses moisture sensitivity and phase separation before finalizing excipient choices.



Surface Chemistry Mapping

Identifies hydrogen bond donors/acceptors, hydrophobic patches, and electrostatic profiles to guide excipient matching.

Computational Methods

Three orthogonal approaches quantify drug-excipient compatibility:

COSMO-RS

Molecular Dynamics

Flory-Huggins χ

Key Parameters

Flory-Huggins χ — predicts polymer-drug miscibility

COSMO-RS ΔH_{int} — quantifies interaction enthalpies

$\chi < 0$ — indicates favorable mixing

Risk Flags

Moisture sensitivity modeled via sorption isotherms; phase separation risk from χ parameter and free energy of mixing (ΔG_{mix}).

Output

Scientifically justified, ranked excipient list ready for experimental confirmation and formulation development.

POC Status

Eight POC tasks spanning image analysis, polymorphic purity, and crystal structure determination are completed or in progress across five API compounds.

S.No	Project	Task	Remarks
1	Resin Beads Nano Particles (SEM)	Image Analysis	Completed — PSD results shared
2	Evocalcet-Formic Acid	Polymorphic Purity	Powder-XRD Indexed; Lattice Parameters determined and refined
3	Evocalcet-Formic Acid	Crystal Structure Determination	Ab initio methods; crystal structure successfully determined
4	Encukalner (Azetukalner)	Polymorphic Purity	Material is Polymorphically impure
5	Voclosporin (Peptide)	Polymorphic Purity	Powder-XRD Indexed; Lattice Parameters determined and refined
6	Tafamidis Form I	Polymorphic Purity	Powder-XRD Indexed; Lattice Parameters determined and refined
7	Tafamidis Form IV	Polymorphic Purity	Powder-XRD Indexed; Lattice Parameters determined and refined
8	Atirmociclib	Polymorphic Purity	Powder-XRD Indexed; Lattice Parameters determined and refined

Image Analysis (Accuracy Test Completed)

Automated morphological image analysis pipeline for crystal habit classification from SEM/optical microscopy — instrument-agnostic, with segmentation as the core technical bottleneck.

Automated Image Analysis

Source agnostic Instrument-Agnostic

Not limited to any particular microscopy source — compatible with SEM and optical microscopy images

Morphological filtering applied to isolate and characterize individual crystal particles

Image-based classification assigns habit categories to segmented crystal regions

Accuracy test completed — pipeline validated on reference datasets

Segmentation Challenges

Core Technical Bottleneck

1  **Overlapping / agglomerated particles** — boundary detection fails when crystals cluster or touch

2  **Irregular shapes** — non-uniform morphologies confound standard contour algorithms

3  **Broken particles** — fragmented crystals produce partial segment outlines misleading classifier

4  **Surface texture** — fine surface features misinterpreted as particle boundaries

SEM Image Analysis — Nano Particles

SEM-based automated image analysis of nano particles reveals a submicron distribution with D50 = 0.37 μm and RSD of 34.63%, consistent with nanoparticle polydispersity.

Median Diameter (D50)

0.37 μm

Submicron Range

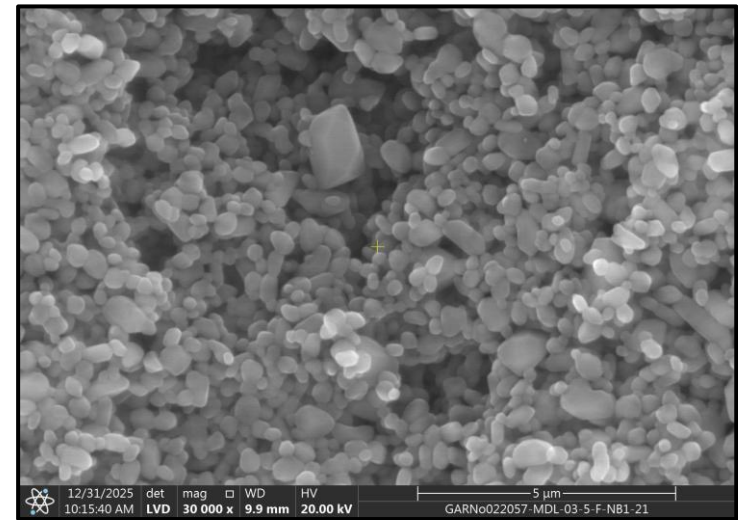
D10 — D90 Range

0.25 — 0.55 μm

RSD — Relative Std Dev

34.63 %

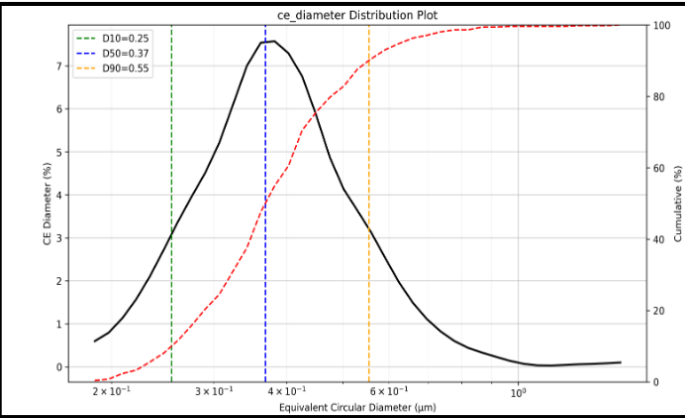
Higher than resin bead RSD (10.22%)



Full PSD Statistics — Nano Particles

Parameter	Value	Parameter	Value
D10	0.25 μm	Mean	0.40 μm
D50	0.37 μm	SD	0.14 μm
D90	0.55 μm	RSD	34.63%
Span	0.82	Max	1.54 μm
D43	0.64 μm	Min	0.18 μm
D32	0.52 μm	SSA	11.43 μm

Higher RSD (34.63%) reflects expected nanoparticle polydispersity — consistent with submicron particle behavior in SEM image analysis.



Particles after segmentation

**Number of particles: 519

CE_Diameter: 1.30μm

CE_Diameter: 0.36μm

CE_Diameter: 0.86μm

CE_Diameter: 0.50μm

CE_Diameter: 0.39μm

CE_Diameter: 0.50μm

CE_Diameter: 0.45μm

CE_Diameter: 0.35μm

CE_Diameter: 0.47μm

CE_Diameter: 0.57μm

CE_Diameter: 0.49μm

CE_Diameter: 0.44μm

CE_Diameter: 0.37μm

CE_Diameter: 0.55μm

CE_Diameter: 0.67μm

CE_Diameter: 0.35μm

CE_Diameter: 0.44μm

CE_Diameter: 0.42μm

CE_Diameter: 0.86μm

CE_Diameter: 0.46μm

CE_Diameter: 0.31μm

CE_Diameter: 0.32μm

CE_Diameter: 0.62μm

CE_Diameter: 0.42μm

CE_Diameter: 0.41μm

CE_Diameter: 0.38μm

CE_Diameter: 0.42μm

CE_Diameter: 0.43μm

CE_Diameter: 0.67μm

CE_Diameter: 0.33μm

Evocalcet Formic Acid — Molecular Structure & PXRD Analysis

Evocalcet Formic Acid ($C_{25}H_{28}N_2O_4$, chiral) was characterized via Powder-XRD indexing using X-Cell with Pawley profile fitting to refine lattice parameters.

Molecular Formula



$C_{25}H_{28}N_2O_4$

Chiral Molecule

Characterization Methodology

- 1 Powder-XRD data collected and processed for indexing
- 2 Unit cell indexed using **X-Cell** algorithm — lattice parameters determined
- 3 Lattice parameters refined using **Pawley profile fitting**

Refined Parameters

Peak Shape
U, V, W

Zero-Point Error
Refined

Method
Pawley Fitting

Indexing Tool
X-Cell

 Only peak shape parameters (U, V, W) and zero-point error were refined at this stage — no structural model parameters included in the Pawley fit.

Evocalcet Formic Acid — Polymorphic Form Purity

PXRD indexing confirms polymorphic purity with excellent Pawley fit ($R_{wp} = 10.46\%$) and refined monoclinic C2 lattice parameters.

Conclusion

Polymorphically Pure

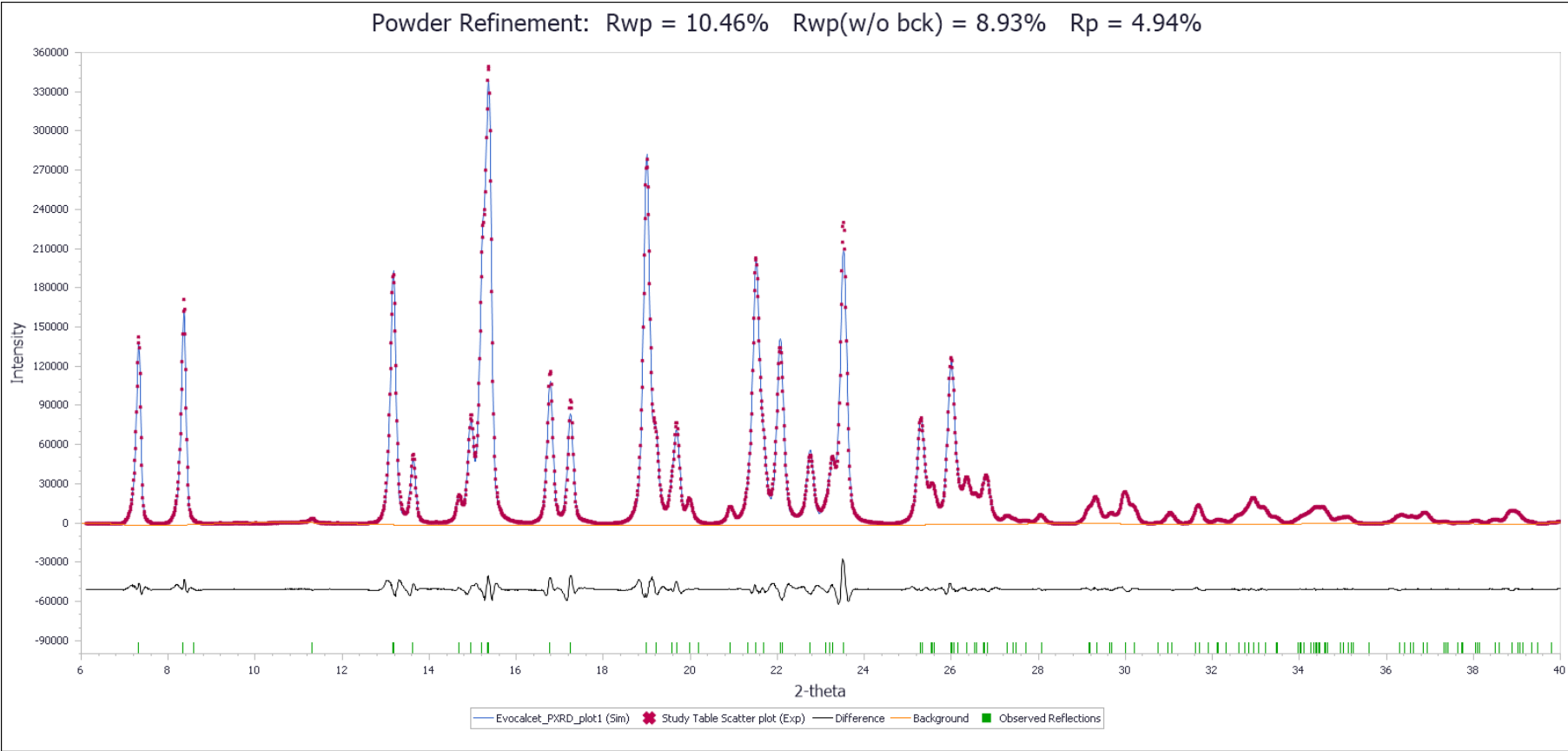
No unindexed peaks detected. Material confirms single polymorphic form.

R_{wp}
Weighted profile factor

10.46%

Refined Lattice Parameters

Parameter	Value
a	26.320 Å
b	8.239 Å
c	11.529 Å
β (beta)	113.953°
Cell Volume	2284.87 Å ³
Z, Z'	4, 1
Space Group	C2 (Monoclinic)
R_{wp}	10.46%



Evocalcet Formic Acid — Conclusions

Powder-XRD analysis conclusively confirms Evocalcet-Formic Acid is polymorphically pure, with excellent indexing and no unindexed peaks.

- Powder-XRD of Evocalcet-Formic Acid **successfully indexed**; lattice parameters determined and refined.
- Low **Rwp = 10.46%** and excellent Pawley difference plot indicate a **high-quality fit**.
- Estimated cell volume slightly lower than experimental ($\sim 2284.87 \text{ \AA}^3$) due to **unknown water stoichiometry**.
- Chiral molecule assigned to centrosymmetric **C2 space group** — chemically reasonable assignment.
- **No unindexed peaks** — excellent agreement between observed and calculated diffraction patterns.

Methodology

- **Ab initio methods** applied for full three-dimensional structure solution
- No single-crystal data required — structure resolved entirely from powder diffraction

Significance

Full 3D structural characterization achieved using only PXRD, demonstrating Aurigene's capability to determine crystal structures of complex chiral APIs from powder data alone.

Polymorphically Pure

Rwp = 11.45%

C2 Space Group



Purity Verdict

Polymorphically Pure

Rwp

10.46%

Excellent Pawley Fit

Space Group

C2

Monoclinic — Chiral

Cell Volume

2284.87 $\text{\AA}^3$

Z / Z' = 4 / 1

Crystal Structure Analysis — Hydrogen Bonding Network

Crystal structure analysis reveals the hydrogen bonding network and molecular packing arrangement of Evocalcet-Formic Acid, confirming the structural model.

✓ Structural Determination Outcome

✓ Crystal Structure Successfully Determined

- Ab initio structure determination from Powder-XRD data
- Full three-dimensional structure resolved without single-crystal requirements
- Evocalcet-Formic Acid — Monoclinic C2 space group confirmed

⚙️ Hydrogen Bonding Network

- Intermolecular hydrogen bonding network characterized between Evocalcet and formic acid moieties
- Proton-transfer interactions analyzed; consistent with salt/co-crystal classification
- H-bond donor/acceptor geometry confirmed via structure refinement

📦 Molecular Packing Arrangement

- Packing arrangement confirmed consistent with C2 space group symmetry
- Chiral molecule in centrosymmetric C2 — chemically reasonable assignment
- $Z = 4$, $Z' = 1$; one molecule in asymmetric unit

📊 Key Structural Parameters

Space Group

C2

Cell Volume

2284.87 Å³

Rwp

10.46%

Crystal System

Monoclinic

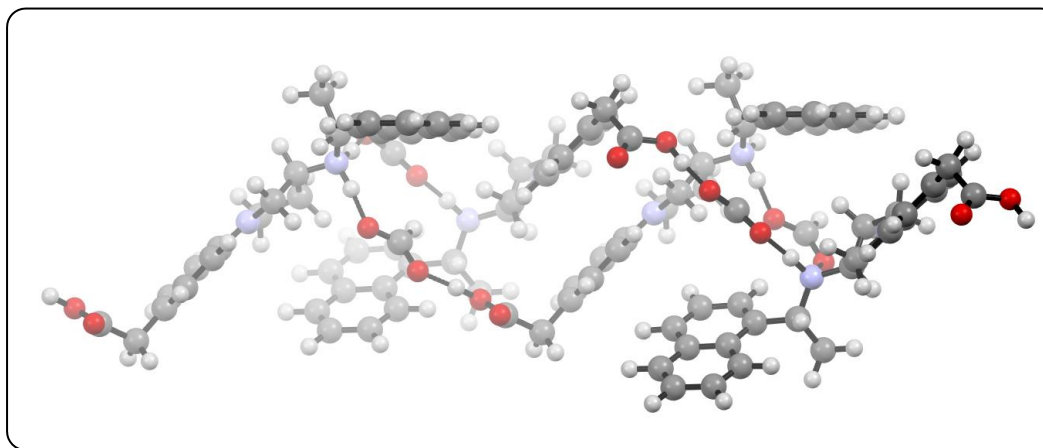
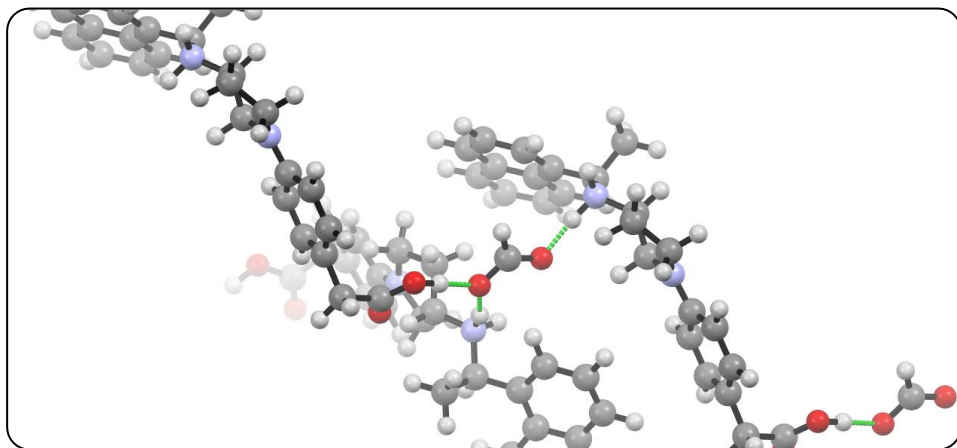
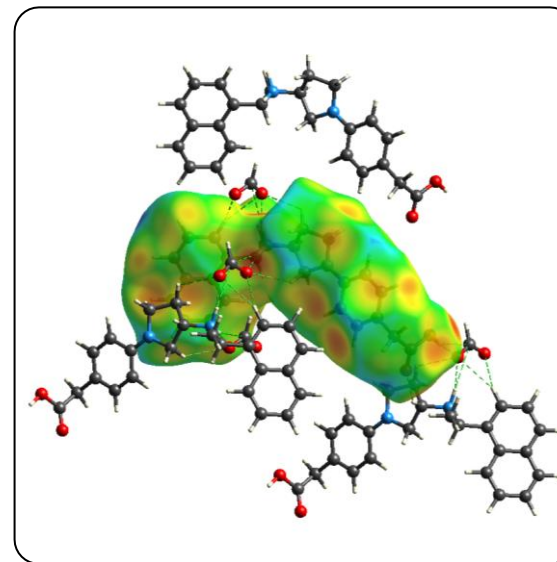
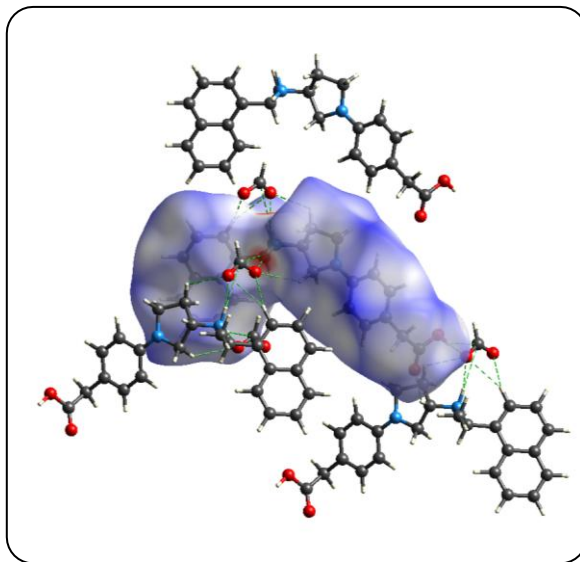
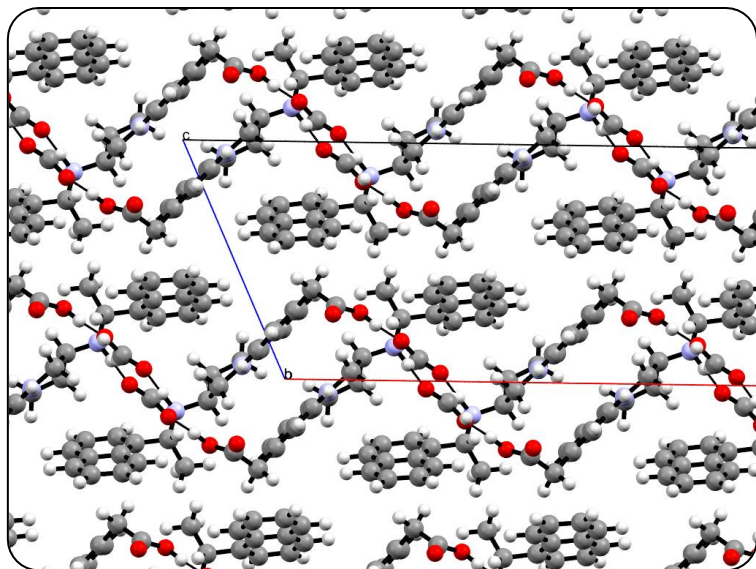
Z / Z'

4 / 1

Crystal structure model fully validated — no unindexed peaks; excellent agreement between experimental and calculated diffraction profiles.

Evocalcet Formic Acid — Crystal Structure Determined from PXRD

The crystal structure of Evocalcet-Formic Acid was successfully determined from Powder-XRD using ab initio methods — enabling full structural characterization without single-crystal data.



Voclosporin — Molecular Structure & PXRD Analysis

Voclosporin ($C_{22}H_{27}ClFN_5O_3$, chiral peptide) characterized via PXRD indexing with Pawley profile fitting; orthorhombic crystal system determined.

Molecular Identity



$C_{22}H_{27}ClFN_5O_3$

 Chiral Molecule



Crystal System

Orthorhombic

Indexing & Profile Fitting Methodology

- 1 **Powder-XRD Indexing** — performed using **X-Cell** algorithm for unit cell determination
- 2 **Pawley Profile Fitting** — full-profile refinement applied to confirm lattice parameters and assess fit quality
- 3 **Crystal System** — orthorhombic symmetry determined from systematic absence analysis

Refined Parameters

Peak Shape — U

Gaussian

Peak Shape — V

Lorentzian

Peak Shape — W

Mixed

Zero-Point Error

Refined

-  Peak shape parameters (U, V, W) and zero-point error refined — no atomic coordinates refined at this stage.

Voclosporin — Polymorphic Form Purity

PXRD indexing of Voclosporin confirms polymorphic purity with excellent Rwp of 4.61% in the orthorhombic $P2_1 2_1 2_1$ space group.

✓ Polymorphically Pure — Confirmed

Key Highlights

Fit Quality (Rwp)

4.61%

Space Group

$P2_1 2_1 2_1$

Crystal System

Orthorhombic

Cell Volume

7228.40 Å³

Z / Z'

4 / 1

Refined Lattice Parameters

Parameter	Value	Unit
a	34.866	Å
b	16.123	Å
c	12.858	Å
Crystal System	Orthorhombic	—
Cell Volume	7228.40	Å ³
Z / Z'	4 / 1	—
Space Group	$P2_1 2_1 2_1$	—
Rwp	4.61%	—

Summary:

Powder-XRD indexed using X-Cell; Pawley profile fitting with peak shape parameters (U, V, W) and zero-point error

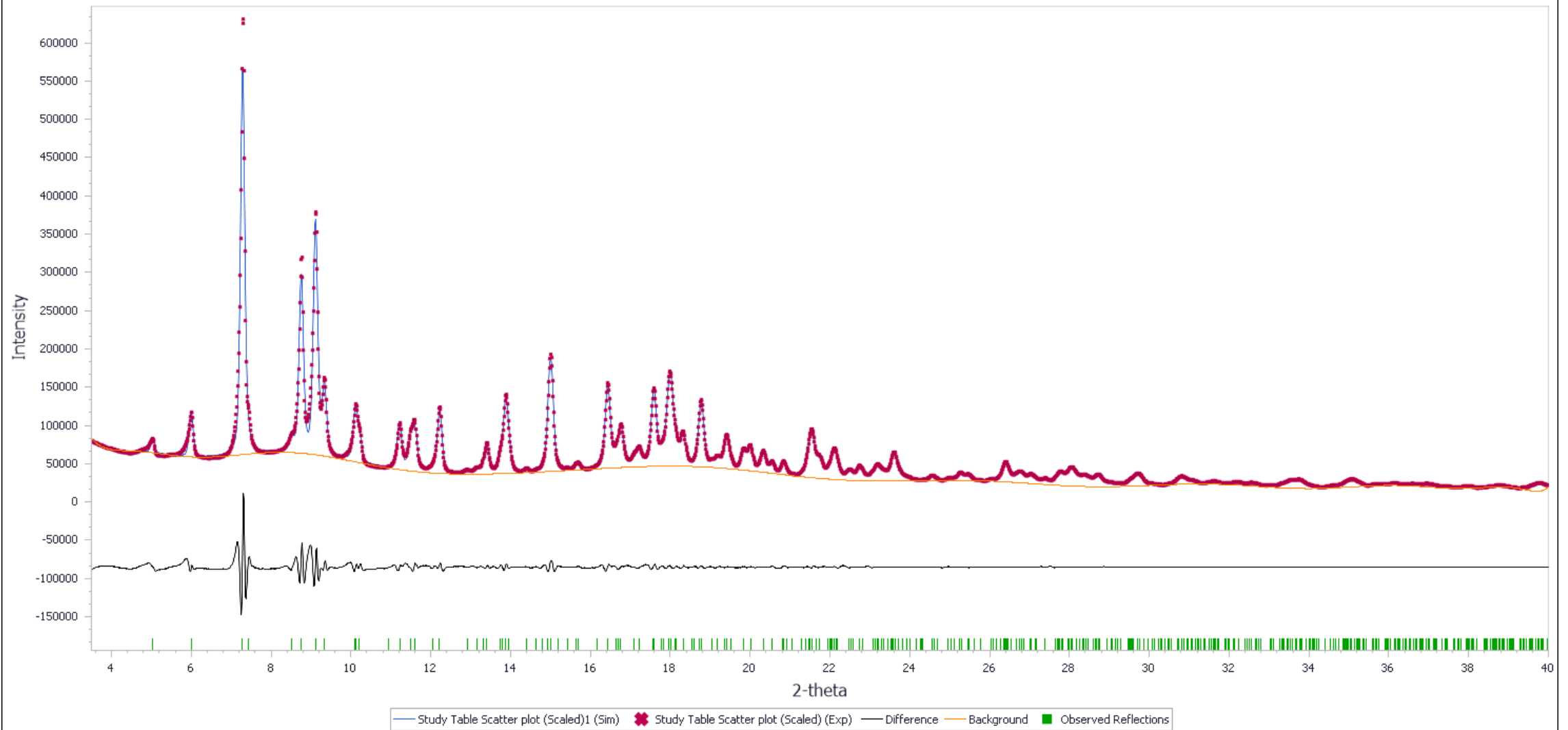
Volume estimate: Estimated volume (~6,880 Å³; hydrate details unavailable) vs. experimental (~7,228.40 Å³) — small discrepancy attributable to unknown hydrate stoichiometry.

Chirality confirmed: Chiral molecule in $P2_1 2_1 2_1$ — non-centrosymmetric space group, chemically appropriate.

No unindexed peaks: All peaks accounted for; excellent agreement between indexed and observed pattern.

Voclosporin — Conclusions

Powder Refinement: $R_{wp} = 4.61\%$ $R_{wp}(w/o\ bck) = 11.21\%$ $R_p = 2.71\%$



Conclusion: Voclosporin is polymorphically pure.

Atirmociclib — Polymorphic Form Purity

PXRD indexing confirms polymorphic purity with $R_{wp} = 3.82\%$ in monoclinic P2 1 with two independent molecules in the asymmetric unit.

 **Polymorphically Pure**

Molecule
Formula **C 22 H 27 ClFN 5 O 3**
Chirality **Chiral molecule**
System **Monoclinic**
Method **X-Cell + Pawley fitting**

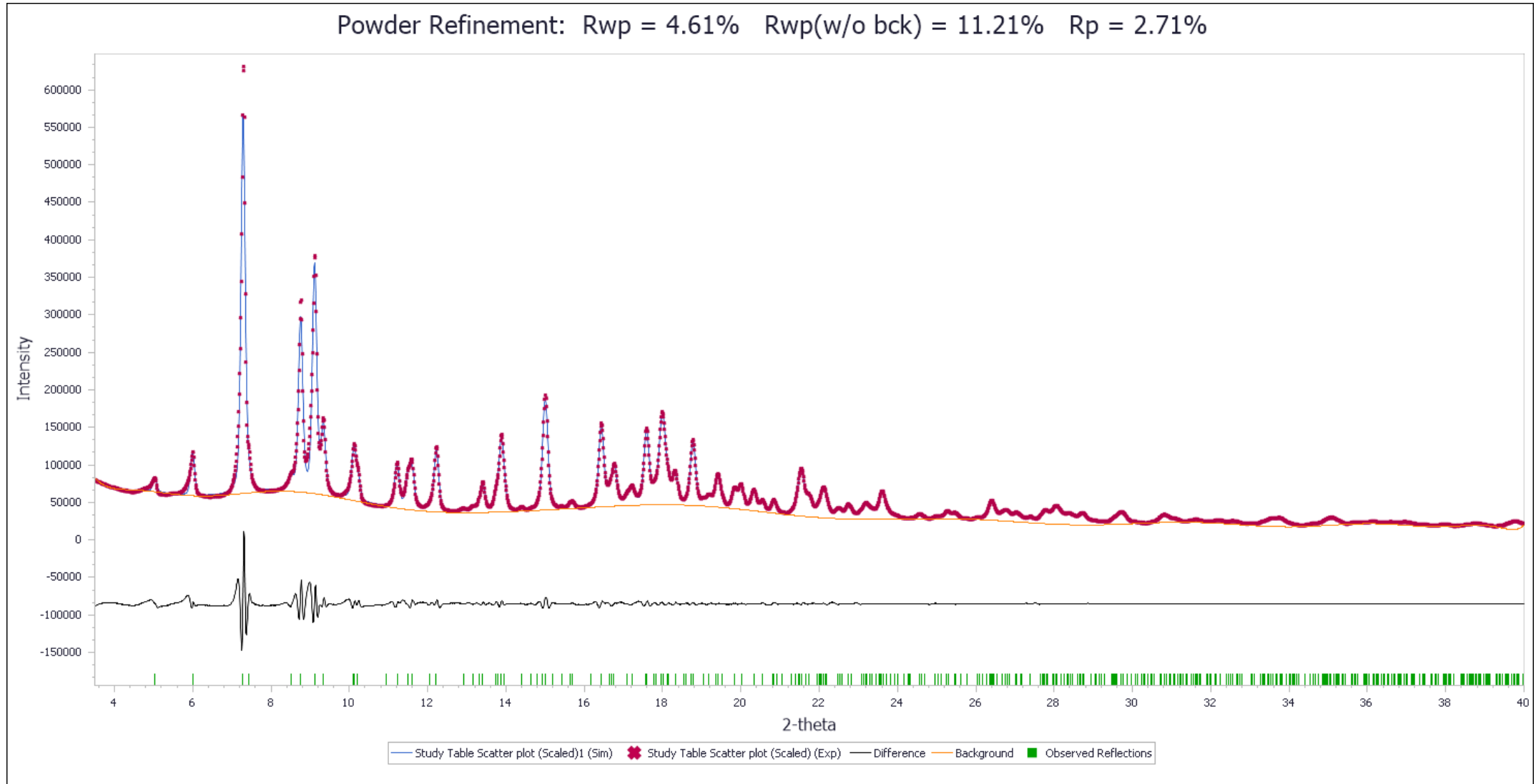
Rwp **3.82%** Excellent fit

 Asymmetric Unit
Z = 8 | Z' = 2 independent molecules

Refined Lattice Parameters

Parameter	Symbol	Refined Value
Axis a	a	12.466 Å
Axis b	b	20.522 Å
Axis c	c	9.876 Å
Monoclinic angle	β	107.589°
Cell Volume	V	2408.76 Å ³
Space Group	SG	P21
Z / Z'	Z, Z'	8 / 2
Profile R-factor	Rwp	3.82%

Atirmociclib — Pawley Profile Fitting



Conclusion: Voclosporin is polymorphically pure.

Atirmociclib — Conclusions

Atirmociclib PXRD analysis confirms polymorphic purity; estimated volume (~2560 Å³) closely matches experimental (2408.76 Å³), validating the monoclinic P2₁ model.

- Powder-XRD of **Atirmociclib successfully indexed**; lattice parameters determined with high confidence.
- **Low Rwp (3.82%)** and excellent Pawley difference plot confirm a high-quality profile fit.
- **Two independent molecules** in asymmetric unit ($Z = 8$, $Z' = 2$); estimated volume ~2560 Å³ closely matches experimental 2408.76 Å³.
- Chiral molecule assigned to **P2₁ space group** — non-centrosymmetric, chemically appropriate for a chiral compound.
- **No unindexed peaks** — all observed diffraction peaks accounted for, excellent agreement with the refined model.

Conclusion

Polymorphically Pure

Atirmociclib is confirmed polymorphically pure — all PXRD evidence consistent with a single monoclinic P2₁ phase.

Space Group	P2 ₁ (Monoclinic)
Rwp	3.82%
Z / Z'	8 / 2
Cell Volume (exp.)	2408.76 Å ³
Est. Volume	~2560 Å ³

| Crystal Structure Determination — Rietveld Refinement

Crystal structure was determined from powder-XRD

Rietveld Refinement — Methodology

1 Ab Initio Structure Input

Crystal structure of Atirmociclib determined from Powder-XRD used as starting model for Rietveld refinement

2 Pattern Fitting

Experimental diffraction pattern fitted against the calculated pattern derived from the structural model

3 Parameter Refinement

Simultaneous refinement of lattice parameters, atomic positions, peak shape, and background

4 Validation

Excellent experimental-calculated agreement confirms ab initio structure determination from Powder-XRD

Refinement Outcome

Excellent Experimental – Calculated Pattern Agreement

Compound

Atirmociclib

Monoclinic $P2_1 \cdot Z/Z' = 8/2$

Structure Source

Ab Initio from Powder-XRD

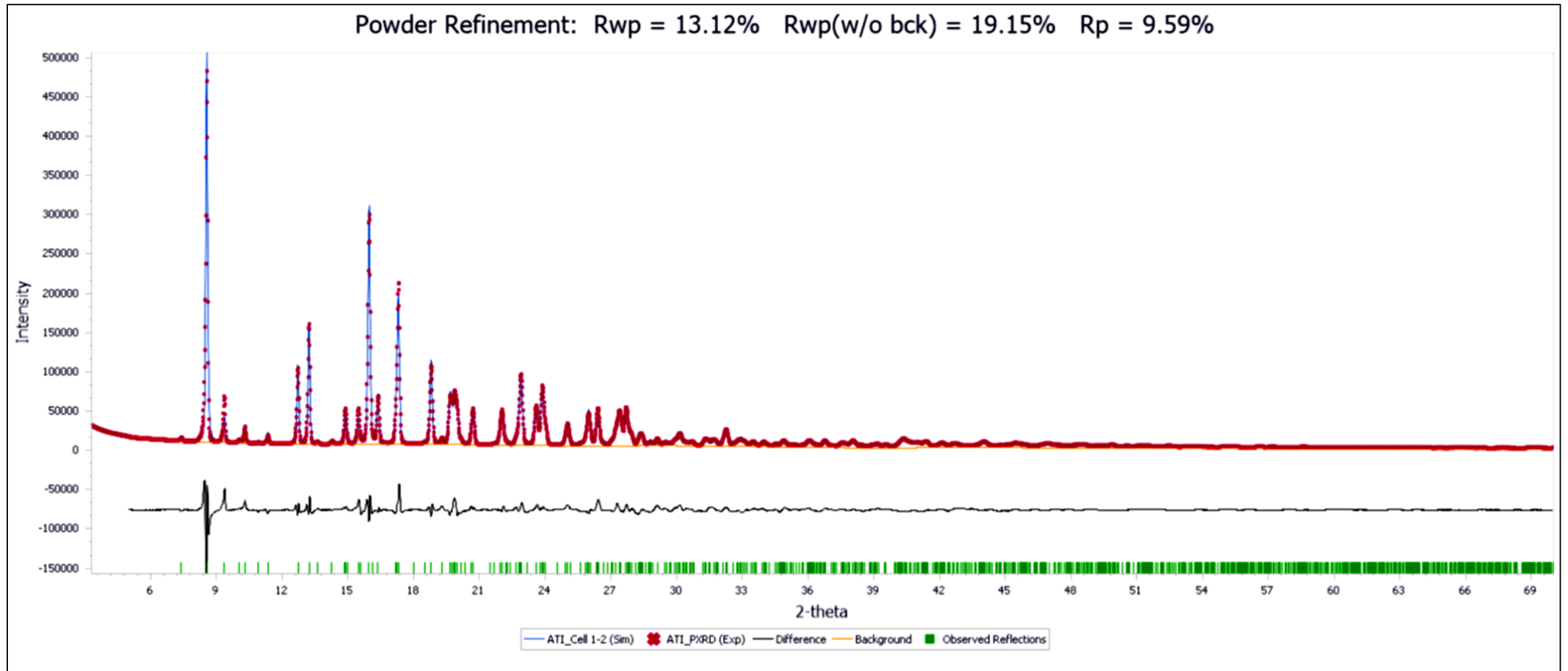
No single-crystal data required

Structural Validation

Crystal structure confirmed

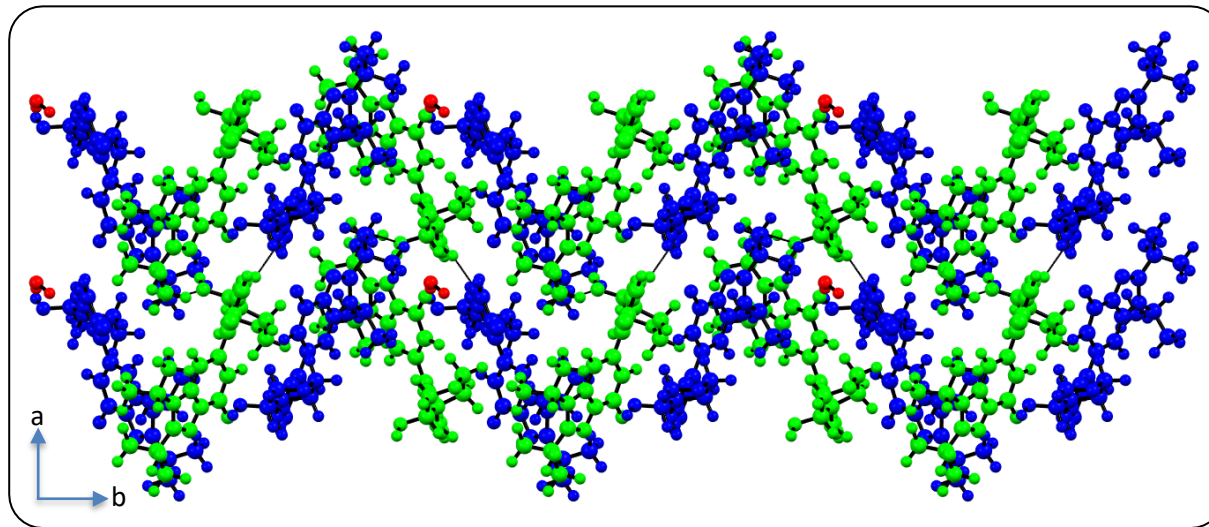
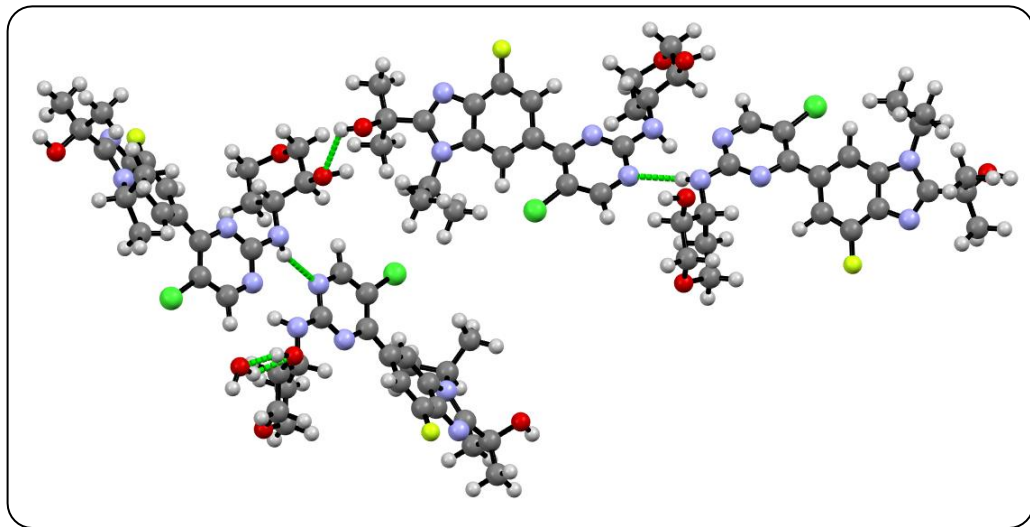
2 independent molecules in asymmetric unit

Atirmociclib — Rietveld Fitting



Conclusion: Atirmociclib is polymorphically pure.

Crystal Structure Determined from Powder-XRD — Atirmociclib

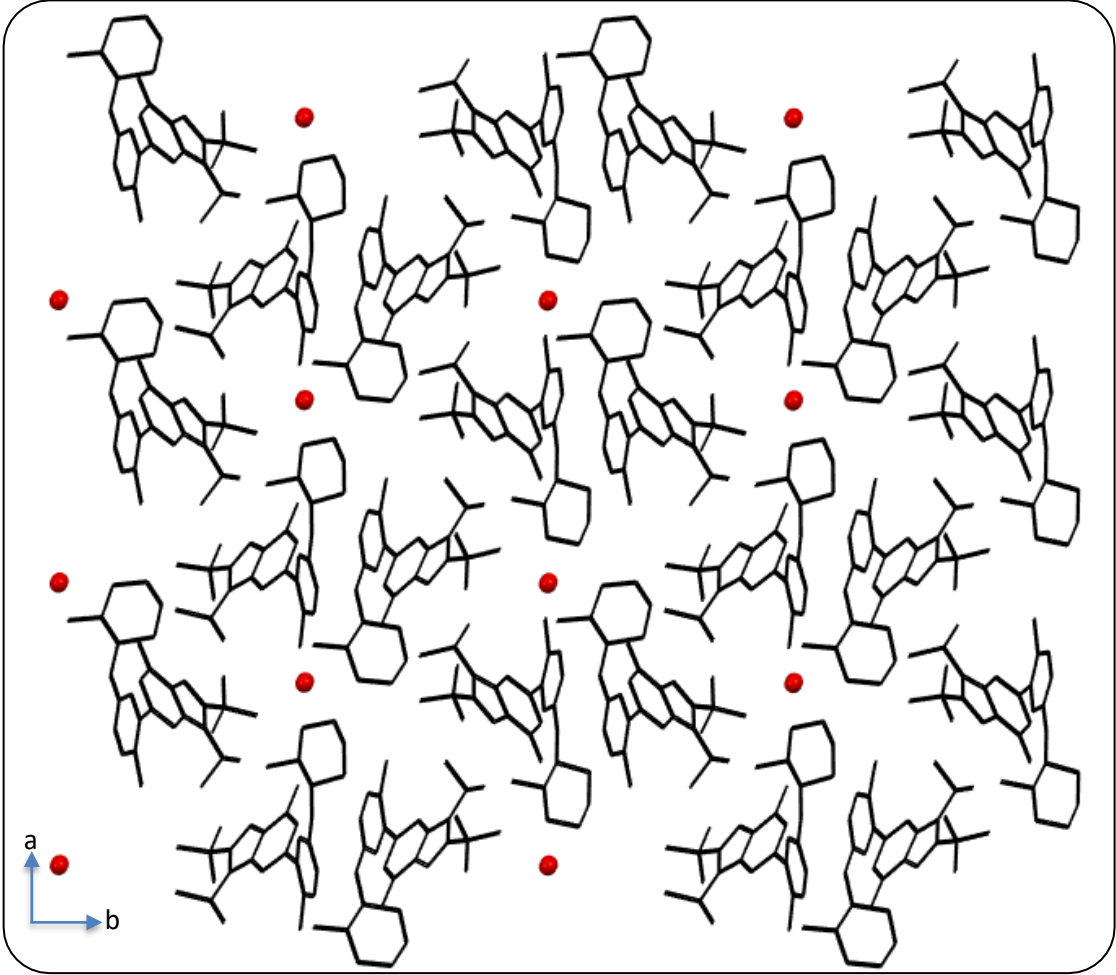
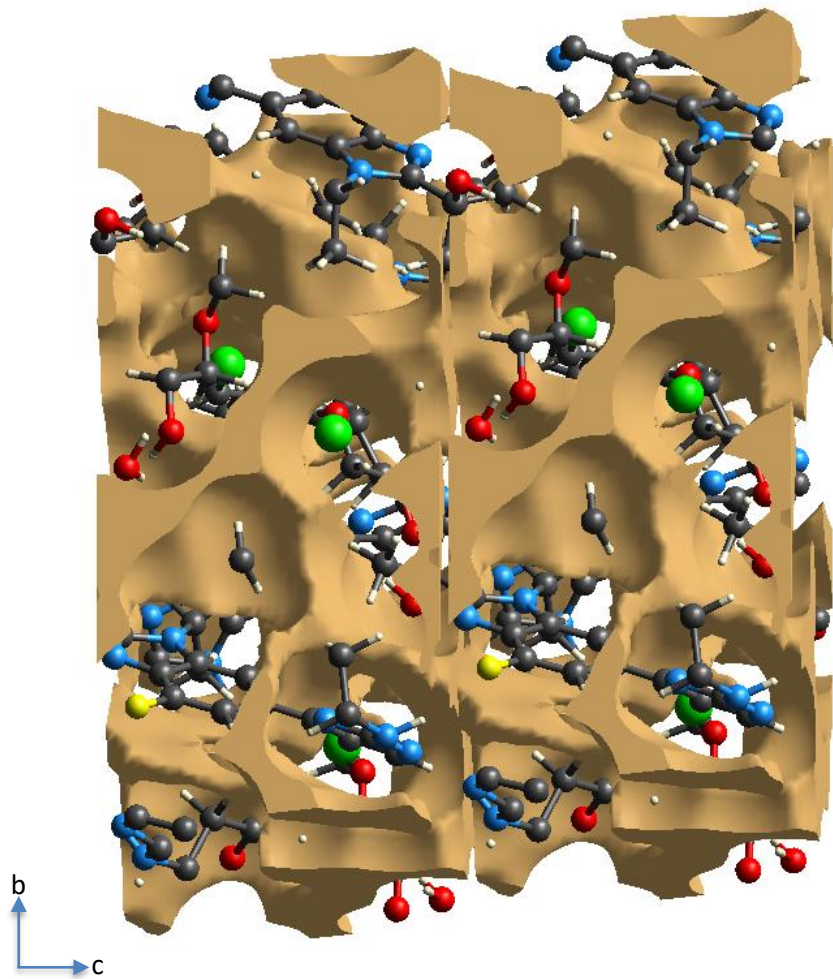


Summary:

O $\cdots$ H-O interaction between Isopropyl groups and N-H...N interactions between amino and pyrimidinyl functional groups forms zig-zag array along

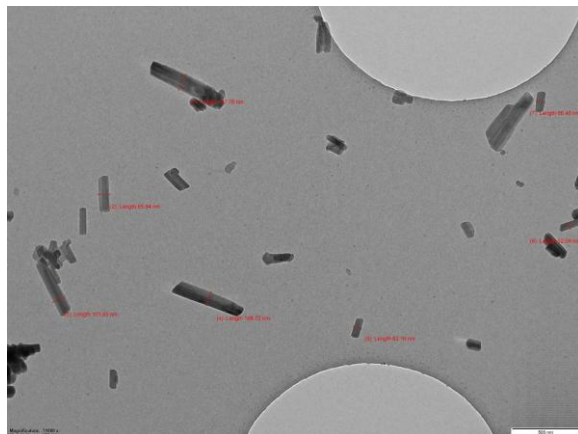
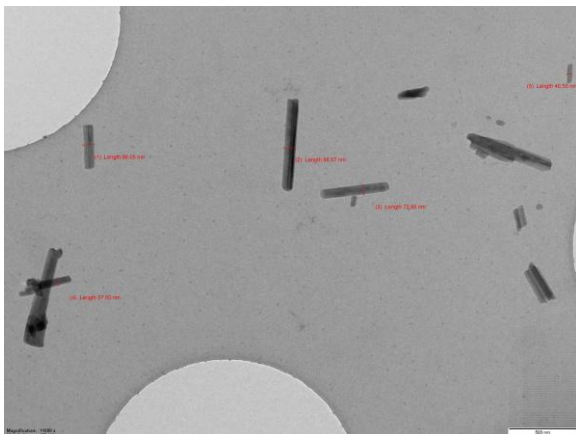
The hydroxyl group on the oxacyclohexane functional group interacts with the water molecule in the lattice

Crystal Structure Determined from Powder-XRD — Atirmociclib



Tafamidis (Form I) — Morphology Studies

Batch: AISH000019



Batch: RTTA00009

